Project #5 : Assessing RNN AI dynamics

- The project focuses on **studying the most classical engram, an Hebbian assembly.** Start from your RNN code with Ampa, Nmda, Gaba-A and Gaba-B currents. Create an Hebbian assembly (HA) formed of n_{HA} of the n excitatory (E) neurons, with mean w_{HA} weight (at existing synapses within the HA). The specific goal here is to parametrically **assess the conditions under which the HA displays functional short-term memory.**

- To do so, build **binary (n_{HA}, w_{HA}) maps** for each of the following criteria qualifying spiking activity in the HA, i.e., the existence of 1) a **spontaneous state** at low spiking frequency f_{Sp} in HA's neurons similar to that of the rest of E neurons (e.g., $f_{Sp} < f_{Sp,nonAH} + 2.5Hz$), 2) an **On transition** to a **self-sustained – persistent – state** with a frequency distinct from f_{Sp}, i.e., at a higher frequency f_P (e.g., $f_P > f_{Sp} + 5Hz$) from the spontaneous state upon a « strong » phasic feed-forward current to HA neurones. (e.g., $I_{On} = 2I_{FF}$, with I_{FF} being the tonic feed-forward current current permanently applied to the whole network), and 3) a back **Off transition** from the persistent to the spontaneous state upon a strong hyperpolarizing current (e.g., $I_{Off} = -2I_{FF}$). You will also build **frequency (n_{HA}, w_{HA}) maps** to keep track of f_{Sp} and f_P.

- A simple scenario can be to successively apply I_{On} and I_{Off} stimulations within the same simulation (during $d_{On} = d_{Off} = 100ms$, e.g.), after at least $d_{Period} = 500ms$ of the preceding period. Spiking frequencies shall be estimated by averaging the frequency across **active** AH neurones (i.e., not silent) during the second halve of the preceding phase (i.e., during $d_{Period}/2$) to get rid of transients.

- The five (n_{HA}, w_{HA}) maps will then be combined to get a **regime map** determining the parameter regions within which the HA displays a a) **silent regime** (black; $f_{Sp} = 0Hz$), b) **spontaneous regime** (green; only the spontaneous state), c) **bistable regime** (orange; i.e., coexisting spontaneous/persistent states), d) **persistent regime** (red; only the persistent state), or e) **saturated regime** (violet; $f_P > 100Hz$). Typically, n_{HA} shall range within $[1, n_E]$ neurons and w_{HA} within $[0,5]\ \mu_w$. Criteria parameters are given as guidelines but shall be adapted if necessary, as well as model parameters, within reasonable ranges to get a HA bistable regime. Be careful also about the initial point taken in the initial (ρ_{EI}, I_{FF}) map.
- The maps will be build using a HA_BATCH.m batch program, similar to what was done in the previous project. Using a 10x10 grid will be a minimum. You shall display a figure with the six maps. You shall also display rasters, as well as exemple of the dynamics of spiking frequency and ionic / synaptic currents (again averaged across the HA) to characterize what occurs in each regime.

- Please send your project before the end of winter vacations.

1) a PDF per binom with a) figures, b) legends and c) a Results section describing the effects observed and their consistency – or not – with the known literature and including explicit references to bibliography and figures;

2) + your Matlab script package.